

Anonymity Testing with Tor and Proxychains

Tools Used

- Tor – Provides anonymity via onion routing and Tor exit nodes.
 - Proxychains-ng – Forces applications to route traffic through Tor (SOCKS5).
 - Curl – Used for IP and traffic verification.
 - Firefox (optional) – For browser-based IP verification.
-

Lab Steps

Step 1: Installing Required Tools

1. Update System Packages

```
sudo apt update
```

2. Install Tor

```
sudo apt install tor -y
```

3. Install Proxychains

```
sudo apt install proxychains -y
```

```
kali-linux-2025.4-vmware-a... Metasploitable2-Linux
root@kali: ~
Session Actions Edit View Help

root@kali:~# sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.1 MB]
Fetched 74.1 MB in 31s (2,379 kB/s)
70 packages can be upgraded. Run 'apt list --upgradable' to see them.

root@kali:~# sudo apt install tor -y
The following packages were automatically installed and are no longer required:
  libstdl2-compat libstdl2-compat-shim
Use 'sudo apt autoremove' to remove them.

Installing:
  tor

Installing dependencies:
  libtorsocks tor-geoipdb torsocks

Suggested packages:
  mixmaster torbrowser-launcher apparmor-utils nylx obfs4proxy

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 70
  Download size: 4,927 kB
  Space needed: 33.3 MB / 52.9 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 libtorsocks amd64 2.5.0-6 [46.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 tor amd64 0.4.9.8-1 [2,085 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 tor-geoipdb all 0.4.9.8-1 [2,768 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 torsocks all 2.5.0-6 [27.9 kB]
Fetched 4,927 kB in 3s (1,610 kB/s)
Selecting previously unselected package libtorsocks:amd64.
(Reading database... 451926 files and directories currently installed.)
Preparing to unpack ./libtorsocks_2.5.0-6_amd64.deb...
Unpacking libtorsocks:amd64 (2.5.0-6)...
Selecting previously unselected package tor.
Preparing to unpack ./tor_0.4.9.8-1_amd64.deb...
Unpacking tor (0.4.9.8-1)...
Selecting previously unselected package tor-geoipdb.
Preparing to unpack ./tor-geoipdb_0.4.9.8-1_all.deb...
Unpacking tor-geoipdb (0.4.9.8-1)...
Selecting previously unselected package torsocks.
Preparing to unpack ./torsocks_2.5.0-6_all.deb...
Unpacking torsocks (2.5.0-6)...
Setting up tor (0.4.9.8-1)...
```

```
root@kali: ~  
Session Actions Edit View Help  
  
(root@kali)-[~]  
└─$ sudo apt install proxychains -y  
The following packages were automatically installed and are no longer required:  
  libssl2-compat libssl2-compat-shim  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
  proxychains  
  
Installing dependencies:  
  libproxychains3  
  
Summary:  
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 70  
  Download size: 22.7 kB  
  Space needed: 74.8 kB / 52.9 GB available  
  
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libproxychains3 amd64 3.1-9+b3 [13.5 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 proxychains all 3.1-9 [9,140 B]  
Fetched 22.7 kB in 13s (1,684 B/s)  
Selecting previously unselected package libproxychains3:amd64.  
(Reading database, 452004 files and directories currently installed.)  
Preparing to unpack ./libproxychains3_3.1-9+b3_amd64.deb...  
Unpacking libproxychains3:amd64 (3.1-9+b3)...  
Selecting previously unselected package proxychains.  
Preparing to unpack ./proxychains_3.1-9_all.deb...  
Unpacking proxychains (3.1-9)...  
Setting up libproxychains3:amd64 (3.1-9+b3)...  
Setting up proxychains (3.1-9)...  
Processing triggers for man-db (2.13.1-1)...  
Processing triggers for kali-menu (2026.2.5)...  
Processing triggers for libc-bin (2.42-16)...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
  
(root@kali)-[~]  
└─$
```

Step 2: Configuring and Starting Tor

1. Start the Tor Service

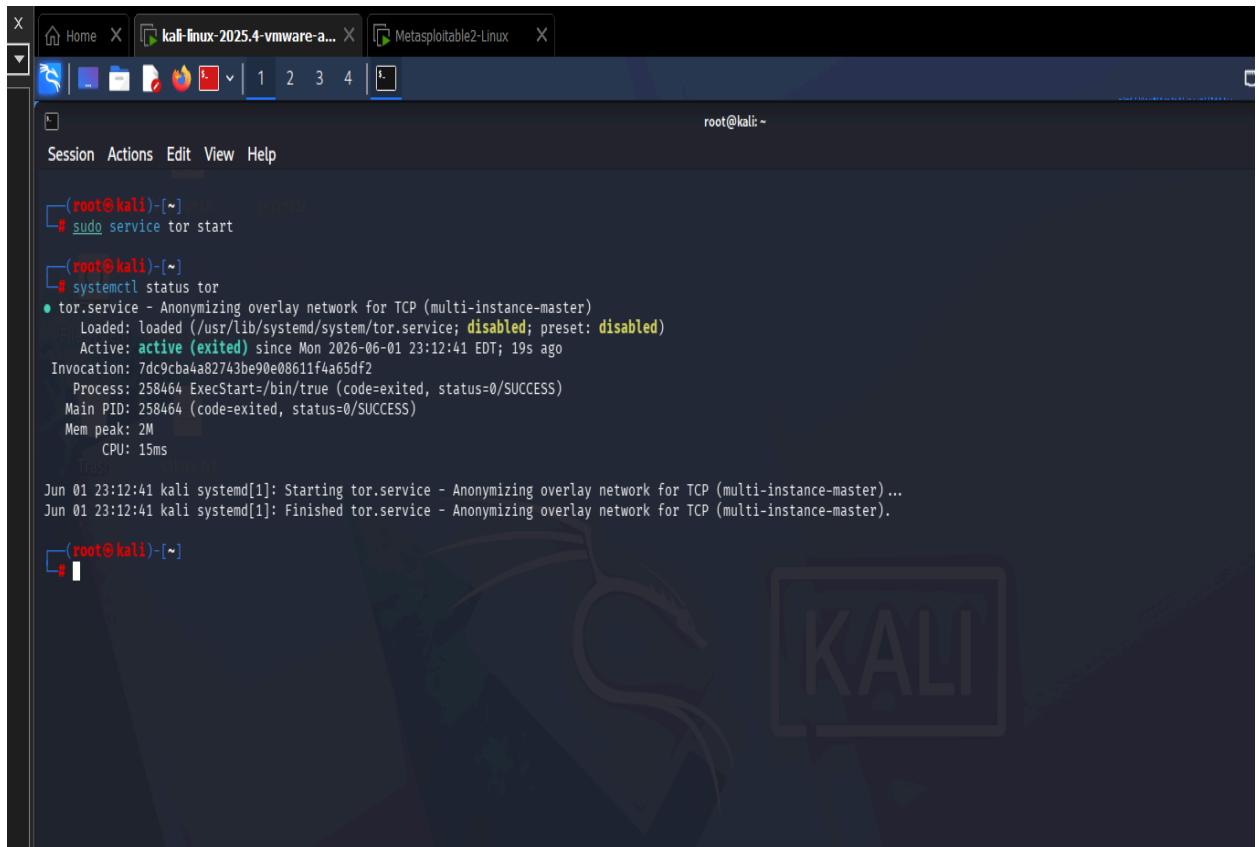
`sudo service tor start`

2. Verify Tor Status

`sudo systemctl status tor`

Exercise 1

- Record the Tor service status output.
- Is the Tor service running successfully?



```
root@kali: ~  
Session Actions Edit View Help  
  
(root@kali)-[~]  
# sudo service tor start  
  
(root@kali)-[~]  
# systemctl status tor  
● tor.service - Anonymizing overlay network for TCP (multi-instance-master)  
   Loaded: loaded (/usr/lib/systemd/system/tor.service; disabled; preset: disabled)  
   Active: active (exited) since Mon 2026-06-01 23:12:41 EDT; 19s ago  
  Invocation: 7dc9cba4a82743be90e08611f4a65df2  
    Process: 258464 ExecStart=/bin/true (code=exited, status=0/SUCCESS)  
   Main PID: 258464 (code=exited, status=0/SUCCESS)  
    Mem peak: 2M  
       CPU: 15ms  
  
Jun 01 23:12:41 kali systemd[1]: Starting tor.service - Anonymizing overlay network for TCP (multi-instance-master) ...  
Jun 01 23:12:41 kali systemd[1]: Finished tor.service - Anonymizing overlay network for TCP (multi-instance-master).  
  
(root@kali)-[~]  
#
```

Tor service status output.

- Loaded: loaded
- Active: active (exited)
- Status: SUCCESS
- Service: tor.service - Anonymizing overlay network for TCP

Is the Tor service running successfully?

Yes. The output shows the service started successfully because the status is active and the process completed with status=0/SUCCESS, indicating no errors occurred during startup.

Step 3: Configuring Proxychains for Anonymity

1. Edit Proxychains Configuration

```
sudo nano /etc/proxychains.conf
```

2. Apply the Following Settings Ensure the configuration includes:

```
strict_chain
```

```
proxy_dns
```

```
tcp_read_time_out 15000
```

```
tcp_connect_time_out 8000
```

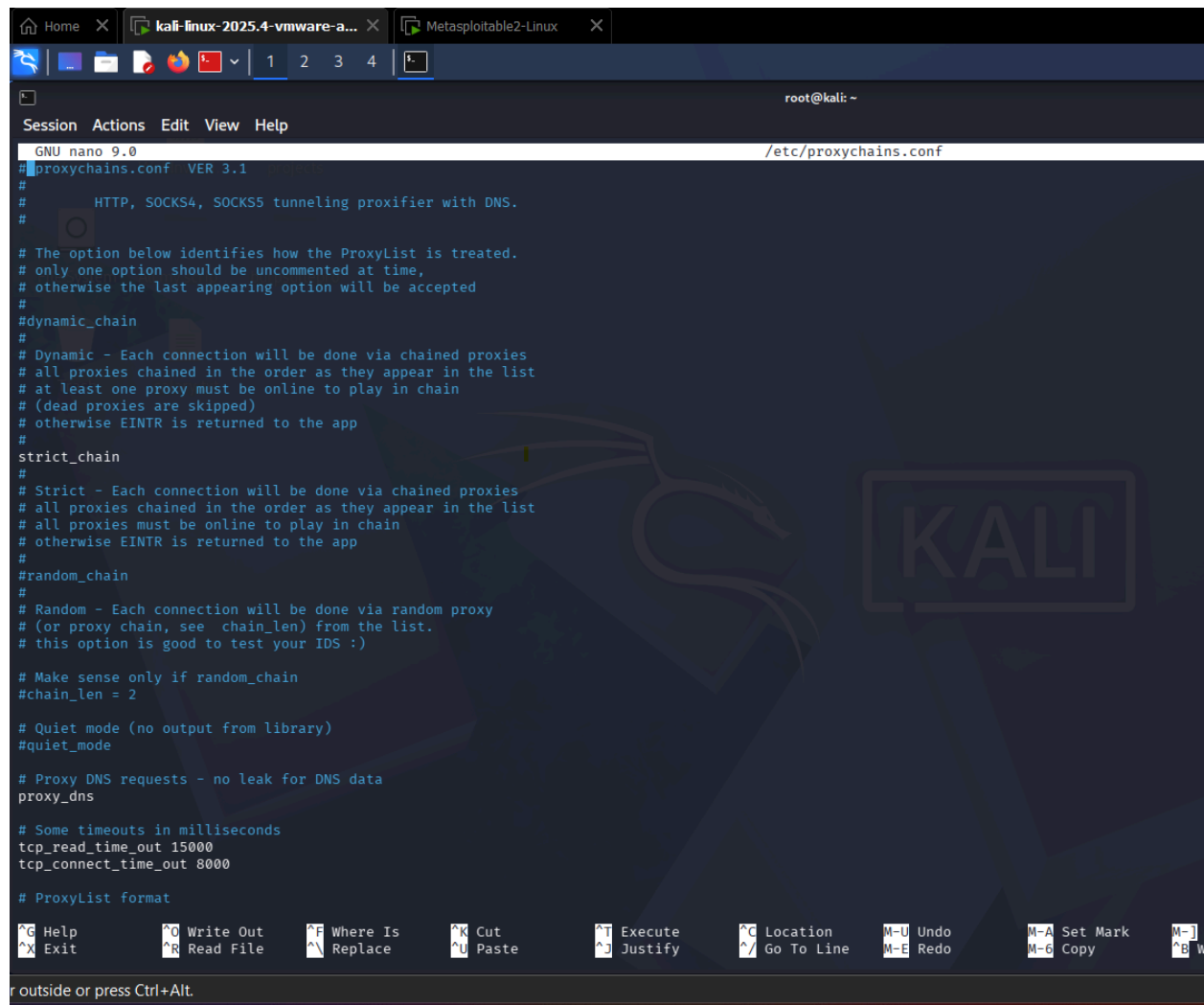
```
[ProxyList]
```

```
socks5 127.0.0.1 9050
```

Note: strict_chain ensures traffic must pass through Tor.

proxy_dns prevents DNS leaks.

Edited Proxychains Configuration



```
GNU nano 9.0 /etc/proxychains.conf
# proxychains.conf  VER 3.1
#
# HTTP, SOCKS4, SOCKS5 tunneling proxifier with DNS.
#
# The option below identifies how the ProxyList is treated.
# only one option should be uncommented at time,
# otherwise the last appearing option will be accepted
#
#dynamic_chain
#
# Dynamic - Each connection will be done via chained proxies
# all proxies chained in the order as they appear in the list
# at least one proxy must be online to play in chain
# (dead proxies are skipped)
# otherwise EINTR is returned to the app
#
#strict_chain
#
# Strict - Each connection will be done via chained proxies
# all proxies chained in the order as they appear in the list
# all proxies must be online to play in chain
# otherwise EINTR is returned to the app
#
#random_chain
#
# Random - Each connection will be done via random proxy
# (or proxy chain, see chain_len) from the list.
# this option is good to test your IDS :)
#
# Make sense only if random_chain
#chain_len = 2
#
# Quiet mode (no output from library)
#quiet_mode
#
# Proxy DNS requests - no leak for DNS data
proxy_dns
#
# Some timeouts in milliseconds
tcp_read_time_out 15000
tcp_connect_time_out 8000
#
# ProxyList format
#
```

Exercise 2

- The difference between strict_chain, dynamic_chain, and random_chain.

Strict_chain requires all proxies to work

Dynamic_chain skips failed proxies, and

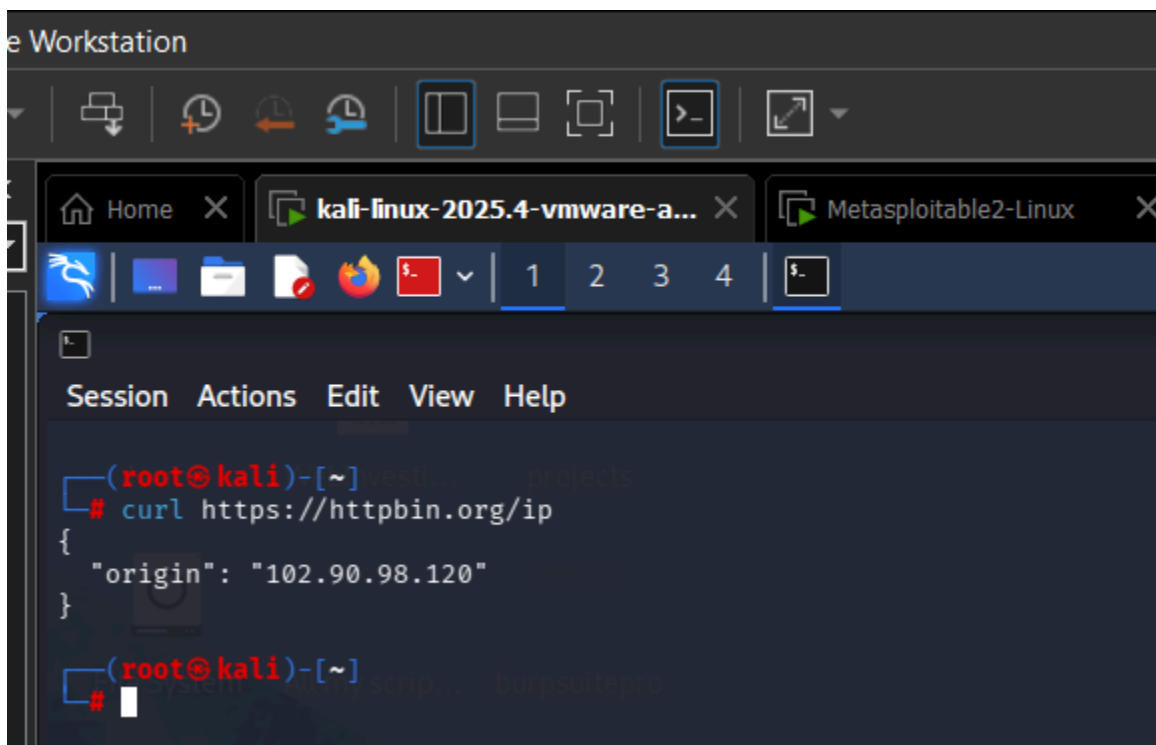
Random_chain chooses proxies randomly for each connection.

Step 4: Baseline IP Address Identification (Non-Anonymous)

1. Check Your Real IP Address

`curl https://httpbin.org/ip`

2. Record the IP Address

A screenshot of a Kali Linux terminal window. The terminal shows the command `curl https://httpbin.org/ip` being executed. The output is a JSON object: `{ "origin": "102.90.98.120" }`. The terminal window has a menu bar with "Session", "Actions", "Edit", "View", and "Help". The terminal prompt is `(root@kali)-[~]`. The terminal window is titled "kali-linux-2025.4-vmware-a..." and is part of a VMware Workstation environment. The terminal window is the active window in the taskbar, which also shows other windows like "Home" and "Metasploitable2-Linux".

origin: 102.90.98.120

Exercise 3

- Is this IP associated with your ISP or local network?

Yes. This is the public IP address assigned by my ISP/network and represents my internet connection.

Step 5: Anonymity Verification Using Proxychains

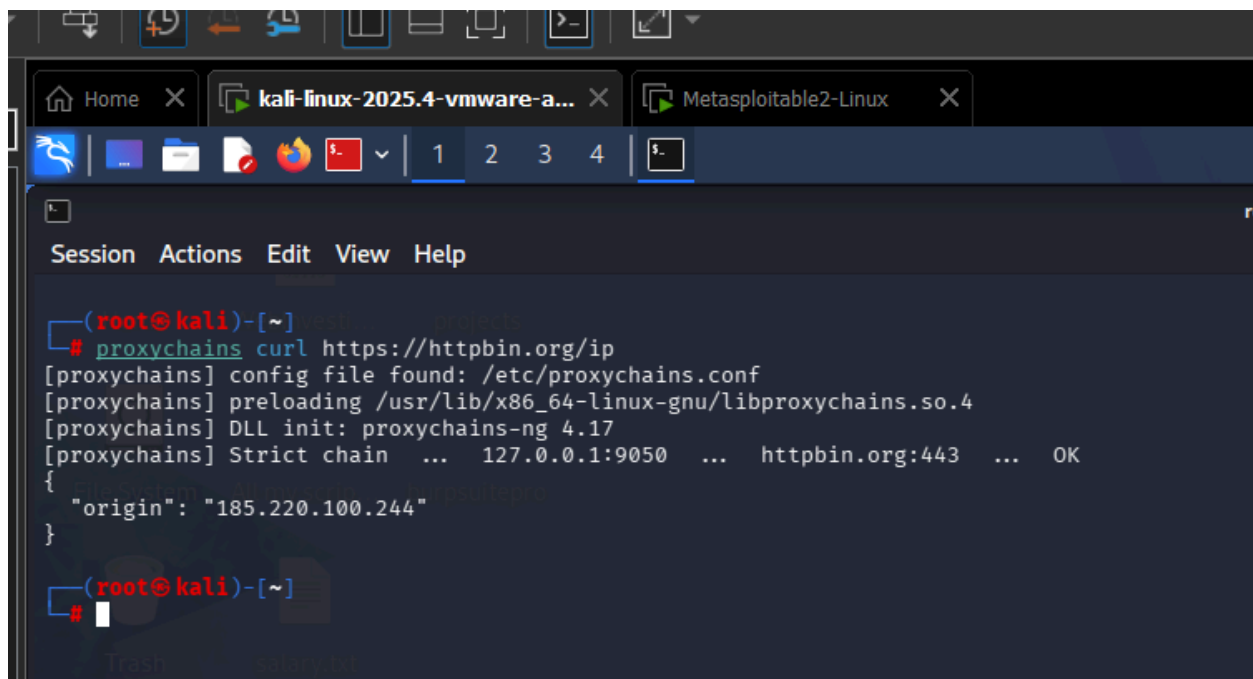
1. Test Anonymous Connection

proxychains curl https://httpbin.org/ip

2. Observe the Output Example:

```
{  
  "origin": "185.183.159.40"  
}
```

Output



The screenshot shows a terminal window with the following output:

```
(root@kali)-[~]  
# proxychains curl https://httpbin.org/ip  
[proxychains] config file found: /etc/proxychains.conf  
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  httpbin.org:443  ...  OK  
{  
  "origin": "185.220.100.244"  
}
```

Exercise 4

- Compare this IP with your real IP from Step 4.
- Is the IP different?

Yes, the IP is different.

- What does this confirm about your anonymity?

It confirms that my traffic is being routed through Tor and the destination website sees the Tor exit node IP instead of my real IP address.

Step 6: Tor Circuit Persistence and IP Behavior

1. Repeat the Command Multiple Times
`proxychains curl https://httpbin.org/ip`
2. Observe Whether the IP Changes

Command repeated Multiple times

```
(kali@kali)-[~]
$ proxychains curl https://httpbin.org/ip

[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:9050 ... httpbin.org:443 ... OK
{
  "origin": "185.181.60.204"
}

(kali@kali)-[~]
$ proxychains curl https://httpbin.org/ip

[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:9050 ... httpbin.org:443 ... OK
{
  "origin": "185.220.101.37"
}

(kali@kali)-[~]
$ proxychains curl https://httpbin.org/ip

[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:9050 ... httpbin.org:443 ... OK
{
  "origin": "185.220.101.37"
}

(kali@kali)-[~]
$
```

Exercise 5

- Did the IP address remain the same or change?
Yes it changed at the beginning but subsequently remained constant when the test was carried out repeatedly.
- What does this indicate about Tor circuit reuse?

The IP remained the same. This indicates Tor reused the existing circuit for multiple connections.

Step 7: Forcing a New Tor Identity (Advanced Test)

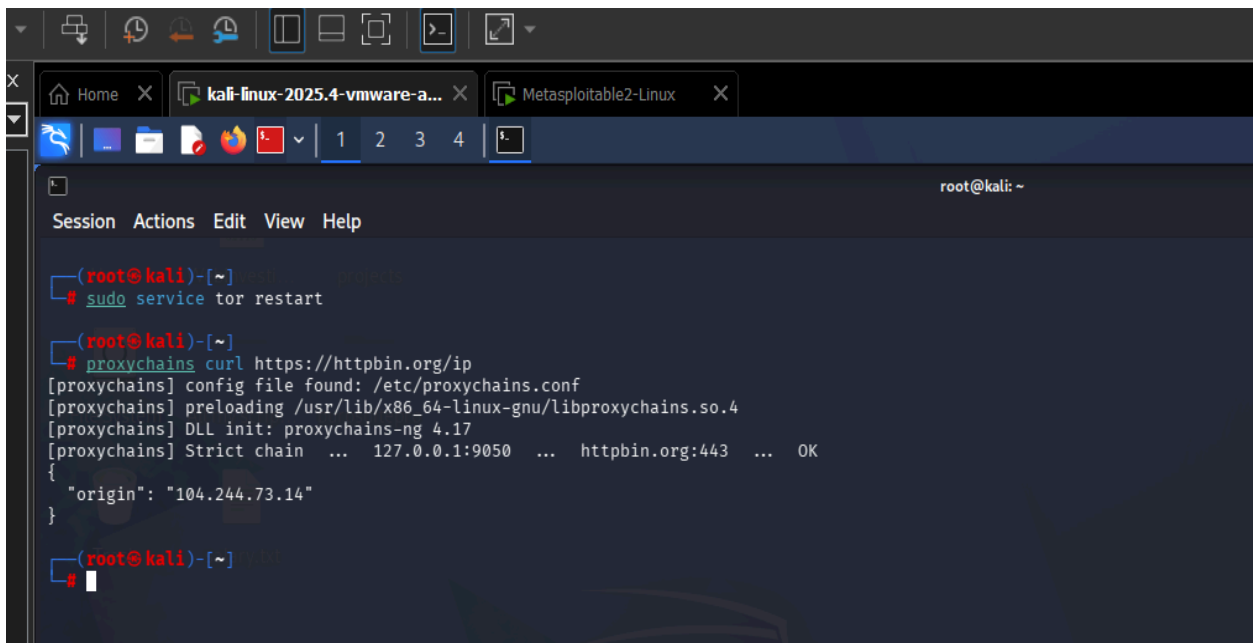
Commands Used:

1. Restart Tor

`sudo service tor restart`

2. Re-test IP Address

`proxychains curl https://httpbin.org/ip`



The screenshot shows a terminal window with the following commands and output:

```
(root@kali)-[~]  
# sudo service tor restart  
  
(root@kali)-[~]  
# proxychains curl https://httpbin.org/ip  
[proxychains] config file found: /etc/proxychains.conf  
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] Strict chain ... 127.0.0.1:9050 ... httpbin.org:443 ... OK  
{  
  "origin": "104.244.73.14"  
}
```

Exercise 6

- Did the Tor exit IP change after restarting Tor?

Yes, the exit IP changed after restarting Tor.

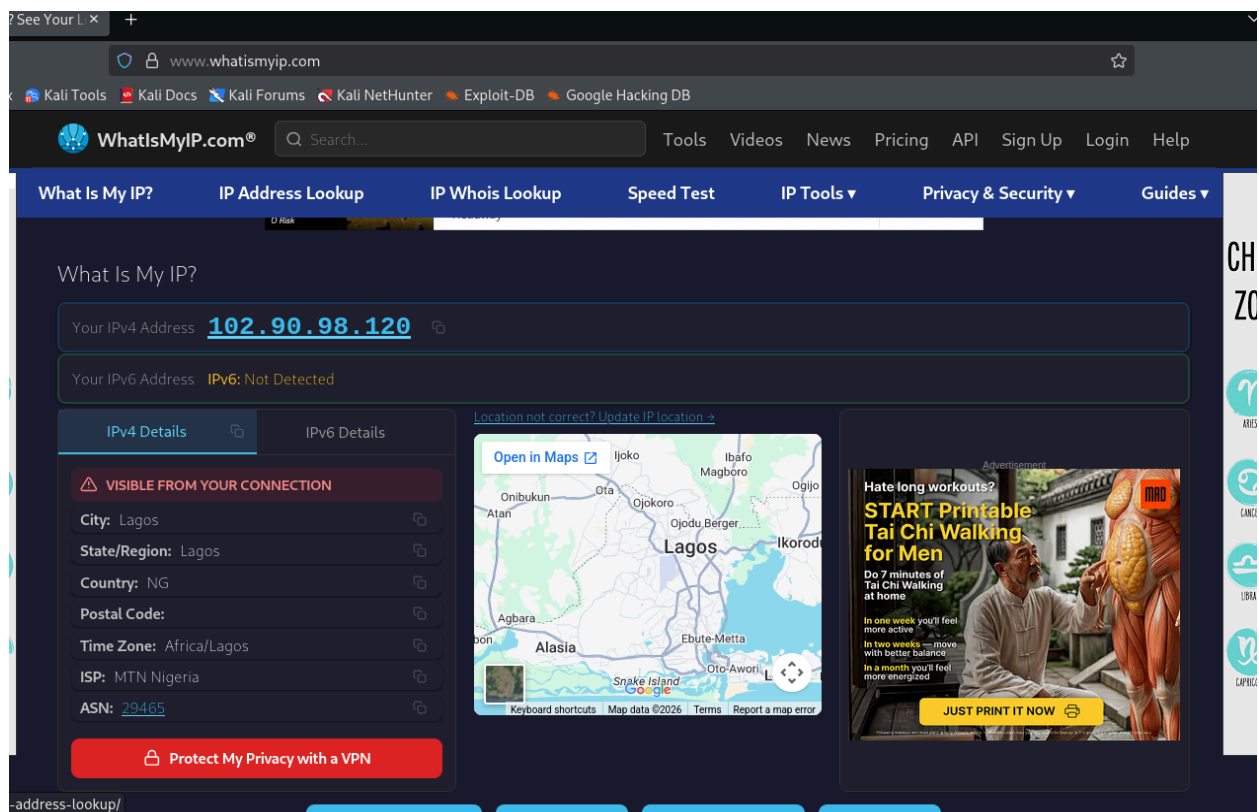
- Why is IP rotation not guaranteed on every request?

Tor reuses circuits for efficiency and anonymity. Restarting Tor may create a new circuit, but the network can still select the same or a similar exit node.

Step 8: Browser-Based Anonymity Testing

Commands Used:

1. Launch Firefox Through Proxychains
proxychains firefox
2. Visit an IP Check Website
 - <https://www.whatismyip.com/>



Exercise 7

- Does the browser show the same Tor exit IP as the curl test?

Yes, the browser shows the same Tor exit IP.

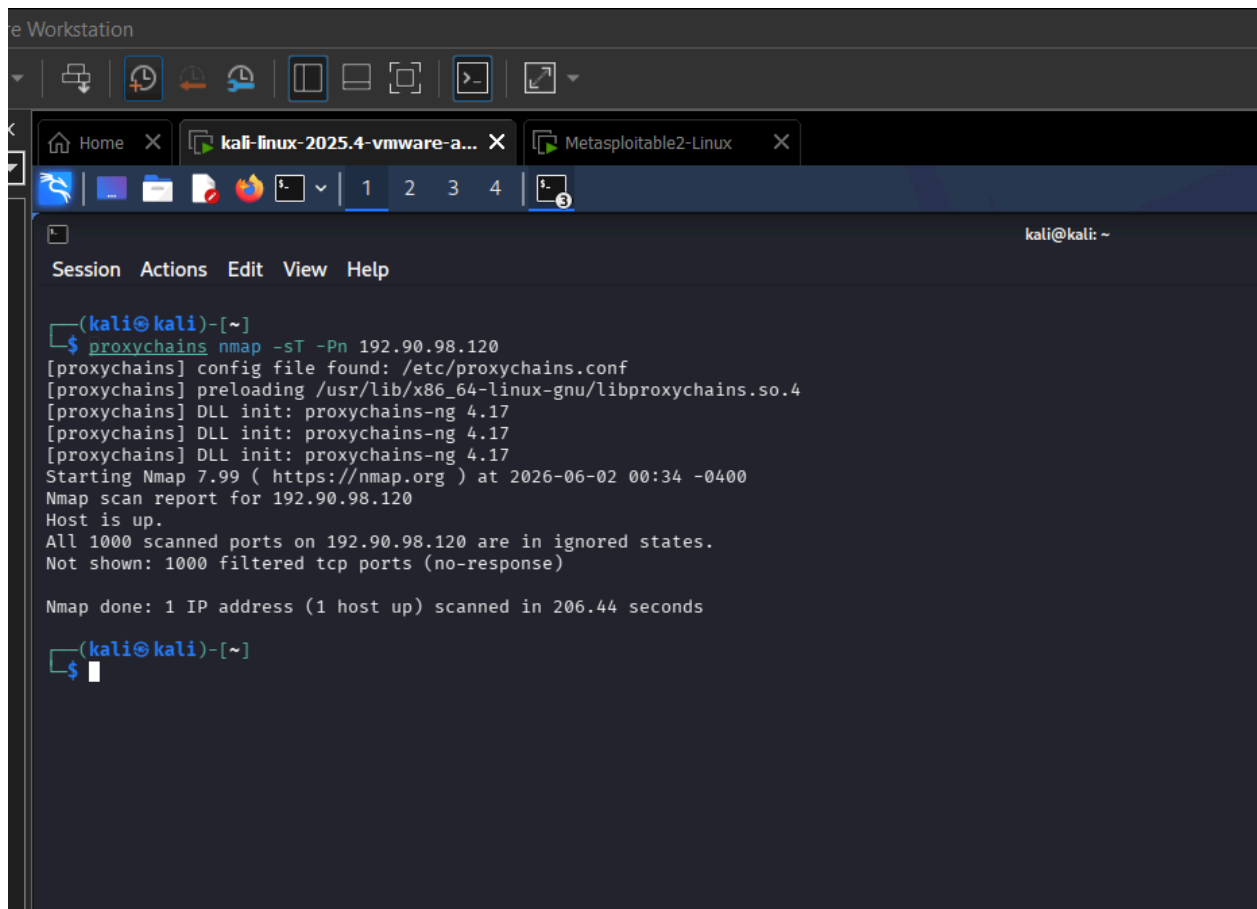
- Why is consistency important in anonymity testing?

Consistency confirms that all traffic is being routed through the intended anonymous path and not leaking through the normal network connection.

Step 9: Anonymity vs. Functionality Trade-Off

1. Optional Tool Test

proxychains nmap -sT -Pn 192.90.98.120



```
(kali@kali)-[~]
$ proxychains nmap -sT -Pn 192.90.98.120
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-02 00:34 -0400
Nmap scan report for 192.90.98.120
Host is up.
All 1000 scanned ports on 192.90.98.120 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 206.44 seconds

(kali@kali)-[~]
$
```

Exercise 8

- What limitations did you experience when routing scans through Tor?

The scan was slower and some connections timed out due to Tor's routing and bandwidth limitations.

- Why is Tor unsuitable for certain types of active scanning?

Tor introduces latency, limits connection types, and many exit nodes block or restrict scanning activity. Large-scale scans can be unreliable and disruptive.

Step 10: Risk and Limitation Analysis

Exercise 9 Answer the following:

- What are the risks of Tor exit nodes?

Traffic leaving the Tor network can be observed by the exit node operator if the connection is not encrypted with HTTPS.

- How can DNS leaks compromise anonymity?

If DNS requests bypass Tor and go directly to the ISP's DNS servers, observers can see which websites are being visited even though the web traffic is routed through Tor.

- Why should Tor not be used for authenticated personal accounts?

Logging into personal accounts can reveal identity through account credentials and activity patterns, reducing or eliminating the anonymity provided by Tor.